

Notes: Shleifer and Vishny (JF, 1992)

Liquidation Values and Debt Capacity: A Market Equilibrium Approach

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1 Big picture

- **Question.** How do **liquidation values** determine firms' **debt capacity**, and why can assets sell far below value in best use when firms are distressed?
- **Setting.** The paper builds a **market-equilibrium theory of asset sales** with three relevant parties:
 - a distressed **seller**,
 - a high-value **industry insider** buyer,
 - a lower-value but less financially constrained **industry outsider**.

The key environment has **aggregate or industry-wide downturns**, so the natural buyers of distressed assets are likely to be financially constrained exactly when the assets come up for sale.

- **Core challenge.** Standard capital-structure theories often take liquidation value as given. This paper asks where liquidation value actually comes from. If the best users of an asset are credit constrained or legally blocked from buying it, then the resale value relevant for debt capacity can be much lower than the asset's value under best management.
- **Main theoretical contribution.** The paper endogenizes liquidation values by focusing on the **financial condition of potential buyers**. Debt overhang that disciplines firms in good states also prevents industry buyers from bidding aggressively in bad states. As a result, liquidation prices become low precisely when liquidation is most likely.
- **Main mechanism.**
 - Firms use debt to prevent managers from undertaking negative-NPV projects.
 - But that same debt creates **debt overhang** in bad states.
 - When a distressed firm must sell assets, the highest-value buyers in the industry may be unable to finance the purchase.
 - The asset is then sold to a lower-value outsider at a **fire-sale price**.
 - Because future liquidation proceeds are low, ex ante debt capacity is lower.
- **Main results.**
 - Asset liquidation need not allocate assets to the highest-value users.
 - Asset illiquidity creates a private cost of leverage and therefore lowers optimal debt levels.
 - **Cyclical assets** and **growth assets** should have lower debt capacity because likely industry buyers are constrained in bad times.
 - **Conglomerates** and **multidivision firms** should have higher debt capacity because they can sell assets more flexibly and in smaller pieces.
 - A firm's optimal leverage depends on the leverage of **other firms in its industry**; the paper introduces the idea of an **industry debt capacity**.
 - Liquidity and debt capacity can be **self-fulfilling**: liquid markets support borrowing, and borrowing in turn supports future liquidity.

- The framework helps explain the **leveraged takeover wave of the 1980s**, when a liquid market for divisions increased debt capacity.
- **Why it matters.** The paper shifts the focus from “How risky is current cash flow?” to “How much cash can an asset raise when sold under distress?” This makes capital structure an equilibrium object that depends on industry conditions, regulation, and business-cycle liquidity.
- **Economic takeaway. Collateral value is not intrinsic to the asset alone.** It depends on who can buy the asset in bad states, how constrained those buyers are, and whether resale markets remain liquid when distress occurs.

2 Data and measurement (key objects)

2.1 This is a theory paper: what the key objects are

- There is **no paper-specific empirical dataset**. The key objects are the seller’s and buyer’s cash flows, debt levels, investment opportunities, liquidation prices, and the identity of the marginal buyer in distress.
- The paper combines:
 - a **formal three-period model** of leverage and liquidation,
 - a set of **economic examples** (farmland, airlines, tankers, corporate divisions), and
 - a **historical application** to the U.S. takeover and leverage boom of the 1980s.

2.2 Main actors in the model

- **Seller.** A firm that may become financially distressed in the bad state and may be forced to liquidate.
- **Industry insider (buyer).** Another firm in the same industry that can operate the seller’s assets better and therefore values them more highly.
- **Industry outsider.** A deep-pocket investor or outsider who values the assets less because of informational and agency disadvantages, but does not face the same debt overhang.

2.3 States, timing, and cash-flow objects

- There are **three periods**: 0, 1, and 2.
 - At date 0, firms choose capital structure.
 - At date 1, the state of the world is realized, firms receive current cash flow, and the seller may face liquidation.
 - At date 2, continuation or post-acquisition cash flows are realized.
- There are **two states**: **prosperity** and **depression**.

- For the **seller**:
 - Y_1^P, Y_2^P : period-1 and period-2 cash flows in prosperity,
 - Y_1^D, Y_2^D : period-1 and period-2 cash flows in depression,
 - I^P : cost of the manager's project in prosperity,
 - R^P : payoff from that project.
- For the **industry buyer**:
 - y_1^P, y_2^P : buyer's cash flows in prosperity,
 - y_1^D, y_2^D : buyer's cash flows in depression,
 - i^P : cost of the buyer's own project in prosperity,
 - r^P : payoff from that project.

2.4 Debt and liquidation values

- The seller issues:
 - short-term debt D_1 due in period 1,
 - senior long-term debt D_2 due in period 2.
- The buyer similarly issues short-term debt d_1 and long-term debt d_2 .
- L^D : the **minimum acceptable liquidation price** for the seller's initial investors.
- l^D : the **maximum price the industry insider can raise** to bid for the seller in depression.
- C_{out} : the period-2 cash flow generated when the outsider owns the seller's assets.
- C_{ins} : the period-2 cash flow generated when the insider owns the seller's assets, with

$$C_{\text{ins}} > C_{\text{out}}.$$

2.5 Why insiders value the assets more

- Industry insiders know the assets better, can run them directly, and avoid some information and agency costs.
- Outsiders face:
 - worse information about asset quality,
 - fear of overpaying,
 - agency costs from hiring specialists to manage the assets.
- So the outsider is usually the **lower-fundamental-valuation buyer**, even though he may be the one who actually wins the auction.

2.6 The paper's motivating example: a distressed farmer

- A distressed farmer may have to sell land.
- The highest-value buyers are neighboring farmers, because they understand the land and can farm it themselves.
- But those neighbors may also be in trouble, or barred from buying, so the land is sold to a deep-pocket outsider who values it less.

Interpretation. This example is the paper's intuition in miniature: distress sales are cheap not because the asset itself becomes useless, but because the best buyers are constrained at the wrong moment.

3 Models and empirical specifications (all equations + interpretation)

3.1 Seller's environment and the agency problem

The seller's manager has a negative-NPV project in prosperity:

$$R^P < I^P. \quad (S1)$$

The seller's period-1 cash flow in depression is low relative to prosperity net of investment:

$$Y_1^D < Y_1^P - I^P. \quad (S2)$$

Overall cash flow is higher in prosperity than in depression even after accounting for the bad investment:

$$Y_1^P + Y_2^P + R^P - I^P > Y_1^D + Y_2^D. \quad (S3)$$

Interpretation.

- The project in prosperity is socially bad but privately attractive to managers.
- Debt is used as a **disciplining device**: it prevents managers from financing the bad project.
- But because depression is sufficiently bad relative to prosperity, a capital structure that blocks bad investment in prosperity may force liquidation in depression.

3.2 How debt prevents bad investment by the seller

To block the bad investment in prosperity, debt must satisfy:

$$I^P > Y_1^P - D_1, \quad (1)$$

so the firm cannot fund the project internally after repaying short-term debt, and

$$I^P - Y_1^P + D_1 > Y_2^P + R^P - D_2, \quad (2)$$

so debt overhang prevents the firm from raising outside funds.

The smallest debt levels satisfying these constraints are:

$$D_1 = Y_1^P - I^P + \varepsilon, \quad (3)$$

$$D_2 = Y_2^P + R^P + \delta. \quad (4)$$

Interpretation.

- Short-term debt forces the firm to come to the market in prosperity.
- Long-term senior debt creates enough debt overhang that new financing is unattractive.
- The capital structure is chosen to solve an agency problem in good times.

3.3 Why the same debt causes liquidation in depression

Given the same debt structure, the seller cannot meet period-1 obligations in depression:

$$Y_1^D < D_1. \quad (5)$$

Nor can it raise enough funds to postpone liquidation:

$$D_1 - Y_1^D > Y_2^D - D_2. \quad (6)$$

If liquidation occurs, initial investors require that avoiding the bad project in prosperity outweigh the loss from liquidation in depression:

$$\Pi^P(I^P - R^P) \geq \Pi^D(Y_2^D - L^D), \quad (7)$$

which implies the minimum acceptable liquidation price

$$L^D = Y_2^D - \frac{\Pi^P}{\Pi^D}(I^P - R^P). \quad (8)$$

Interpretation.

- The more valuable it is to block managerial overinvestment, the more debt investors are willing to tolerate.
- The lower the expected liquidation price, the less attractive debt becomes.
- This is the paper's first core point: **liquidation value directly determines debt capacity.**

3.4 Buyer's environment and debt overhang

The industry buyer faces the analogous agency problem in prosperity:

$$r^P < i^P. \quad (B1)$$

The buyer is less badly hit in depression, but still cannot finance a large purchase from internal funds alone:

$$0 < y_1^D - y_1^P + i^P < C_{\text{out}}. \quad (\text{B2})$$

And prosperity is still overall better than depression:

$$y_1^P + y_2^P + r^P - i^P > y_1^D + y_2^D. \quad (\text{B3})$$

To prevent the buyer's own bad project in prosperity, its debt must satisfy

$$i^P > y_1^P - d_1, \quad (9)$$

$$i^P - y_1^P + d_1 > y_2^P + r^P - d_2, \quad (10)$$

with minimal debt levels

$$d_1 = y_1^P - i^P + \varepsilon, \quad (11)$$

$$d_2 = y_2^P + r^P + \delta. \quad (12)$$

Interpretation.

- The insider buyer is subject to the same capital-structure logic as the seller.
- Debt that disciplines the buyer ex ante also limits its ability to exploit acquisition opportunities ex post.

3.5 Maximum insider bid and endogenous fire-sale pricing

Let l^D denote the maximum amount the insider can raise to buy the seller in depression. The insider can bid at most

$$l^D < (y_1^D + y_2^D) - (y_1^P + y_2^P + r^P - i^P) + C_{\text{ins}}. \quad (13)$$

The buyer's initial owners are willing to impose debt overhang only if the gains from preventing the bad project exceed the losses from not obtaining the liquidating firm cheaply in prosperity:

$$\Pi^P(i^P - r^P) > \Pi^D(C_{\text{ins}} - \text{liquidation price}). \quad (14)$$

Interpretation.

- Even though the insider values the seller's assets more, debt overhang may keep him from raising enough cash to bid that value.
- This is the paper's central equilibrium insight: **the highest-value buyer can lose because he is financially constrained.**

3.6 Liquidation equilibrium

There are two main cases.

Case 1: Insider wins. If

$$l^D > C_{\text{out}},$$

then the insider can outbid the outsider, and the asset goes to the higher-value buyer at price C_{out} .

Case 2: Outsider wins. If

$$l^D < C_{\text{out}},$$

but Condition (14) still holds, the outsider wins the auction even though

$$C_{\text{ins}} > C_{\text{out}}.$$

The liquidation price is then l^D , the maximum amount the insider can pay.

Interpretation.

- Liquidation does **not** necessarily allocate assets efficiently.
- Auction prices do **not** necessarily reflect the second-highest fundamental valuation.
- The liquidation price is pinned down by the financing capacity of the constrained insider, not by the asset's best-use value.

3.7 Industry debt capacity and multiple equilibria

The paper shows that some parameter values generate **two equilibria**.

- In one equilibrium, the **buyer** has high debt overhang and the seller chooses low debt to avoid being liquidated too cheaply.
- In the other equilibrium, the **seller** has high debt and gets liquidated in depression, while the buyer stays unlevered so it can acquire the seller's assets.

Interpretation.

- Individual firms' leverage depends on the expected leverage of other firms in the industry.
- This creates the idea of an **industry debt capacity**: the industry as a whole can sustain only so much leverage because someone must remain financially flexible enough to buy distressed assets.

3.8 Why firms may choose asset sales over rescheduling or new securities

The paper argues that asset sales can dominate other distress-resolution methods because they:

- reduce the asset-substitution problem by repaying debt,
- weaken information asymmetries relative to broad external financing,
- transfer control of the asset to the buyer and therefore reduce agency problems,
- lessen the need for costly coordination among multiple creditors,
- allow firms to raise cash by selling high-value but low-current-cash-flow assets.

Interpretation. Asset sales are often chosen not because they are cheap in an absolute sense, but because **rescheduling and new issuance can be even more costly**.

3.9 Cross-sectional predictions for debt capacity

The model yields several financing predictions.

- **Liquid assets** should carry more debt.
- **Growth assets** should carry less debt because likely insiders have little cash relative to asset value.
- **Cyclical assets** should carry less debt because likely insiders are also distressed in recessions.
- **Smaller firms** may carry more debt within industry because they are easier for others to acquire.
- **Conglomerates and multidivision firms** should carry more debt because they can sell assets selectively and in smaller pieces.

Interpretation. Debt capacity depends not only on volatility or tangibility, but also on the thickness and financial strength of the resale market.

3.10 Time variation, high-market liquidity, and self-fulfilling debt capacity

The paper argues that liquidity varies over time because:

- industry cash flows vary over the business cycle,
- regulation changes the number of potential buyers,
- beliefs about resale conditions affect current willingness to borrow and lend.

The key self-reinforcing logic is:

higher expected resale liquidity → more borrowing today → more future buyers with financing → higher realized liquidity

Interpretation.

- High markets tend to be liquid markets because values and financing capacity rise together.
- Liquidity creates debt capacity, but debt capacity also creates liquidity.
- This opens the possibility of multiple equilibrium asset markets: an illiquid low-debt equilibrium and a liquid high-debt equilibrium.

3.11 Application to the takeover wave of the 1980s

The paper uses the theory to interpret the U.S. 1980s takeover boom.

- Asset acquisitions are observed to be **procyclical**, which is hard to explain if assets always sell at fundamental value.
- In the 1980s, liquidity in the market for corporate divisions rose because of:

- strong corporate cash flows,
 - more buyers,
 - relaxed antitrust enforcement,
 - foreign acquirers,
 - financial innovation.
- That increase in liquidity raised debt capacity and made highly leveraged transactions feasible.

Interpretation. The paper reverses the usual story. Rather than saying junk bonds alone created the takeover wave, it argues that a liquid resale market for assets helped make large debt issues sustainable in the first place.

4 Identification and instruments (what makes the estimates credible)

4.1 What makes the mechanism work

- The paper is not an empirical identification exercise. The key question is which assumptions are doing the work.
- The mechanism requires four ingredients:
 - managers with negative-NPV investment opportunities in good states,
 - debt overhang as a commitment device,
 - industry insiders as the highest-value users of the distressed assets,
 - aggregate or industry-wide bad states that constrain those insiders when liquidation occurs.

4.2 Why aggregate shocks are essential

- If the seller suffers only an **idiosyncratic** shock, other firms in the industry are likely to be healthy and can bid up the asset price.
- The paper's fire-sale result is strongest when the shock is **industry-wide** or **economy-wide**, because then the natural buyers are distressed at the same time.

Bottom line. The model is really about **correlated distress among natural buyers**.

4.3 Why the outsider can win despite lower valuation

- The outsider is less efficient, but does not face the same debt overhang.
- The insider values the asset more, but cannot fully monetize that value through borrowing.
- So the allocation is determined by **pledgeable financing capacity**, not just by technological efficiency.

4.4 Why liquidation value is endogenous here

- In earlier capital-structure models, liquidation value is often a primitive.
- Here, liquidation value depends on:
 - who is allowed to bid,
 - who has cash or borrowing capacity,
 - how well outsiders can operate the assets,
 - whether beliefs sustain liquid resale markets.

Interpretation. The paper’s main novelty is not “distress is costly” but rather **why distress is costly in equilibrium**.

4.5 Why the auction view can fail

- Standard legal arguments for immediate auction rely on the idea that auctions allocate assets efficiently.
- This paper shows that when the best buyers are financially constrained or regulated out, the auction price may reflect the balance-sheet strength of second-best buyers rather than the true value in best use.

4.6 What would undo the result

- The fire-sale logic weakens if:
 - industry buyers are unconstrained,
 - shocks are purely idiosyncratic,
 - many deep-pocket outsiders can run the assets well,
 - debt can be renegotiated costlessly,
 - regulation does not limit insider participation.

4.7 Relation to the literature

- Relative to **Williamson (1988)**, the paper goes beyond redeployability and studies specialized assets with low alternative-use value.
- Relative to **Myers (1977)** and **Hart (1991)**, the paper keeps debt overhang but endogenizes the liquidation value that debt capacity depends on.
- Relative to legal views favoring immediate auction, it shows that auctions need not produce efficient ownership or fair valuations in distress.

5 Tables and figures

5.1 Figure 1: Timing for the seller

- Figure 1 lays out the seller's timing:
 - choose capital structure at date 0,
 - in prosperity, receive cash flow and face the bad investment opportunity,
 - in depression, face low cash flow and possible liquidation,
 - at date 2, either continue or realize liquidation value.
- The figure is important because it shows the paper's key tradeoff: debt solves an agency problem in prosperity but generates liquidation risk in depression.

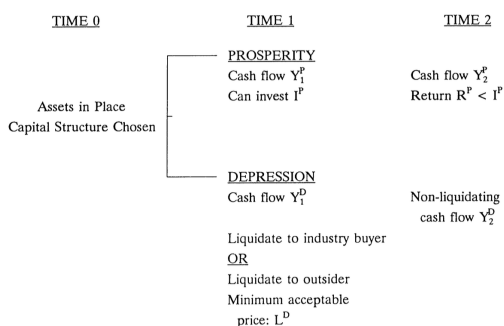


Figure 1. Timing for the seller. R is the future payoff from the investment, I is the cost of the investment, and Y is cash flow from existing assets. Superscripts denote states (either prosperity or depression) and subscripts denote time periods.

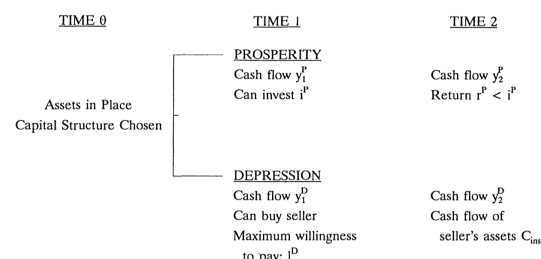


Figure 2. Timing for the industry buyer. r is the future payoff from the investment, i is the cost of the investment, and y is cash flow from existing assets. Superscripts denote states (either prosperity or depression) and subscripts denote time periods.

5.2 Figure 2: Timing for the industry buyer

- Figure 2 mirrors Figure 1 for the insider buyer.
- It highlights the buyer's key role in the mechanism: the same debt structure that disciplines it in prosperity can prevent it from buying the seller in depression.
- This figure is the cleanest visual statement of why the model is a **general-equilibrium theory of liquidation values** rather than a one-firm capital-structure model.

5.3 No formal empirical tables in the paper

- The paper is primarily theoretical and does not contain the kind of regression tables common in modern empirical finance.
- Instead, its “evidence” comes from:
 - comparative statics,
 - equilibrium logic,
 - historical and industry examples.

5.4 The distressed farmer example

- This is the paper's pedagogical example.
- Neighbors are the high-value buyers because they know the land and can operate it directly.
- But if neighbors are constrained or excluded, the farm must be sold to a deep-pocket outsider at a lower price.
- Read this example as the stripped-down intuition behind the formal model.

5.5 Airlines and tankers

- The airline example distinguishes **idiosyncratic distress** from **industry-wide distress**.
 - When only one airline is weak, other airlines can buy its assets.
 - When the whole industry is weak, even natural buyers are constrained and asset prices fall.
- The tanker example shows that distressed assets may be sold cheaply to outsiders, who can then earn very large returns if they have the capital to wait.

5.6 The 1980s takeover wave

- This discussion is the paper's major application.
- Read it as a macro-finance interpretation of takeover activity:
 - liquid division markets increased expected resale values,
 - higher expected resale values increased debt capacity,
 - higher debt capacity supported LBOs and other leveraged transactions,
 - later declines in liquidity helped bring the boom to an end.

6 Short summary

- **Method.** Build a three-period capital-structure model with a distressed seller, a constrained high-value insider buyer, and an unconstrained lower-value outsider.
- **Mechanism.** Debt overhang that prevents bad investment in prosperity also weakens industry buyers in depression, so distressed assets may sell below best-use value.
- **Main implication.** Debt capacity depends on liquidation value, and liquidation value depends on the financial strength of likely buyers, not just on the asset's technology.
- **Predictions.** Cyclical and growth assets have lower debt capacity; conglomerates and multidivision firms have higher debt capacity; liquidity and leverage move together over time.
- **Application.** The paper interprets the 1980s leverage boom as a period when a liquid market for corporate assets raised expected resale values and therefore made much higher leverage feasible.

7 Conclusion

7.1 Core conclusion (what the paper establishes)

- The paper establishes that asset liquidation values are **endogenous market-equilibrium objects**.
- Distress sales need not allocate assets to the highest-value users, because the natural buyers may be financially constrained or legally excluded precisely when assets come up for sale.
- As a result, optimal leverage is limited by **asset illiquidity**, not just by the volatility of operating cash flow.

7.2 Economic intuition

- Debt is useful because it disciplines managers.
- But debt also reduces financial flexibility in bad states.
- If all the likely specialist buyers are similarly levered, then the market for distressed assets becomes thin exactly when firms most need it.
- That makes collateral values low in recessions and raises the ex ante cost of leverage.

7.3 Takeaway

This paper is a classic because it shows that **the value of collateral depends on the state of the market for collateral**. A firm's borrowing capacity is therefore tied not only to its own prospects, but also to the balance sheets, regulations, and expectations governing the buyers who might someday purchase its assets. In that sense, leverage is an equilibrium property of an entire market, not just a single firm.